kasetsart university sriracha campus

MACHINE LEARNING

# LOAN APPROVAL CLASSIFICATION

# DATASET

ชุดข้อมูลประกอบด้วยบันทึก 45,000 รายการและตัวแปร 14 ตัว ซึ่งแต่ละตัวอธิบายไว้ด้านล่าง:

| Column | Description | Type |
|---|---|---|
| person_age | Age of the person | Float |
| person_gender | Gender of the person | Categorical |
| person_education | Highest education level | Categorical |
| person_income | Annual income | Float |
| person_emp_exp | Years of employment experience | Integer |

# DATASET

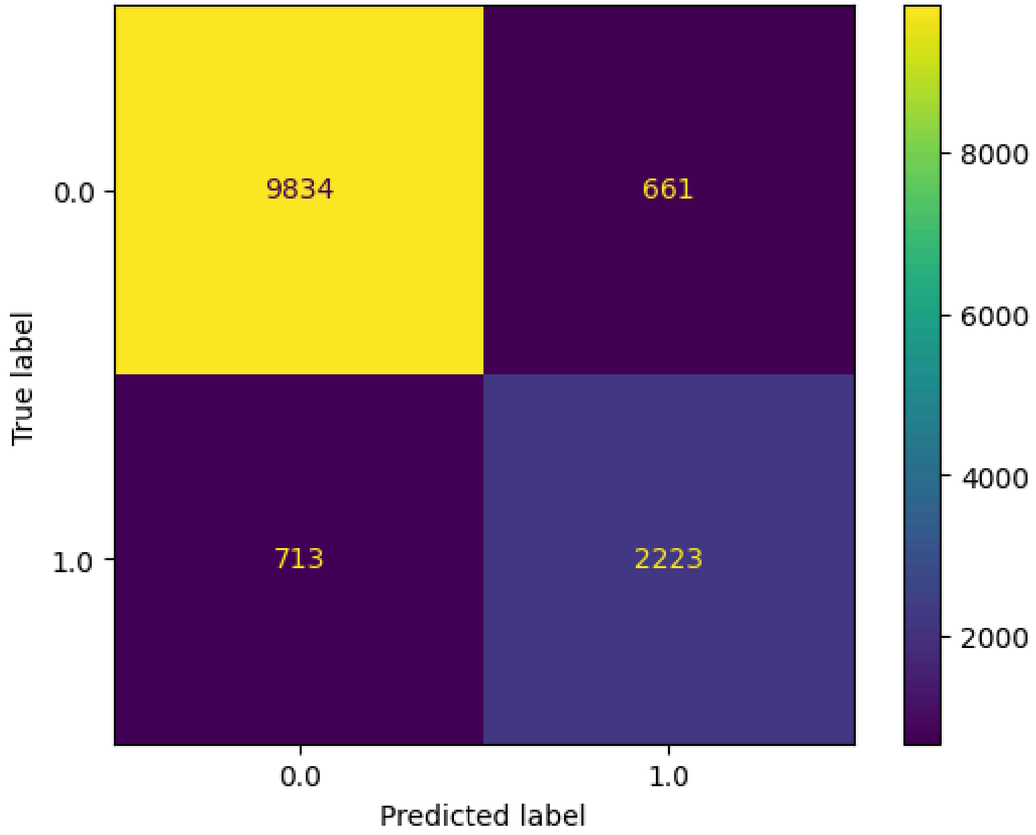
| Column | Description | Type |
|---|---|---|
| person_home_ownership | Home ownership status (e.g., rent, own, mortgage) | Categorical |
| loan_amnt | Loan amount requested | Float |
| loan_intent | Purpose of the loan | Categorical |
| loan_int_rate | Loan interest rate | Float |
| loan_percent_income | Loan amount as a percentage of annual income | Float |

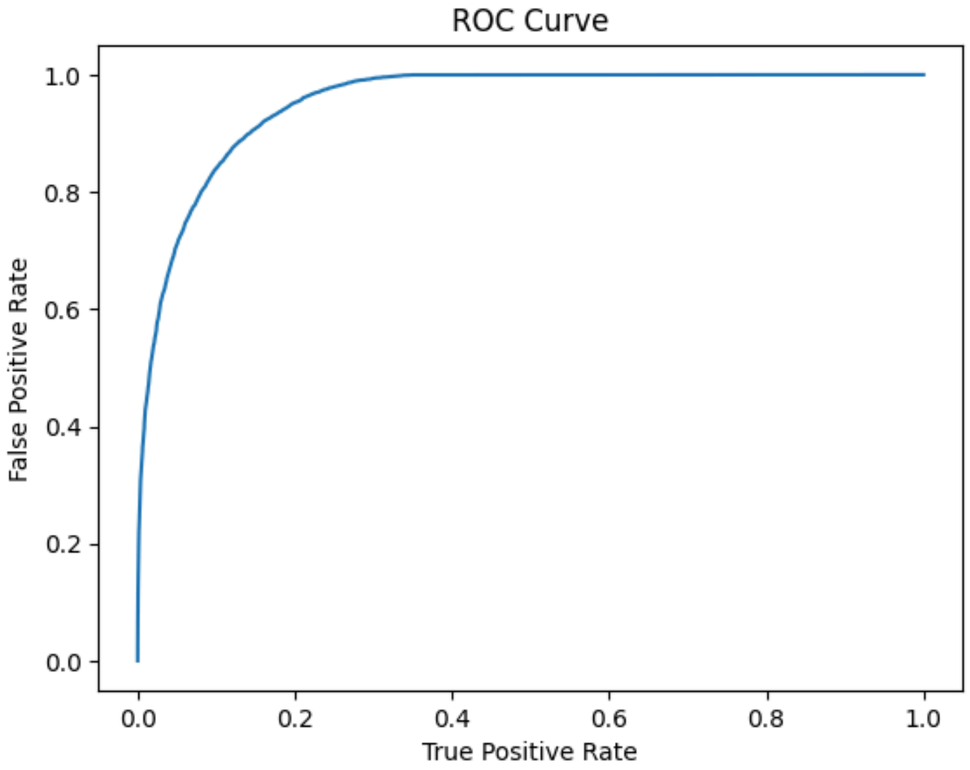
# DATASET

| Column | Description | Type |
|---|---|---|
| cb_person_cred_hist_length | Length of credit history in years | Float |
| credit_score | Credit score of the person | Integer |
| previous_loan_defaults_on_file | Indicator of previous loan defaults | Categorical |
| loan_status (target variable) | Loan approval status: 1 = approved; 0 = rejected | Loan approval status: 1 = approved; 0 = rejected |

# PERFORMANCE: LOGISTIC REGRESSION

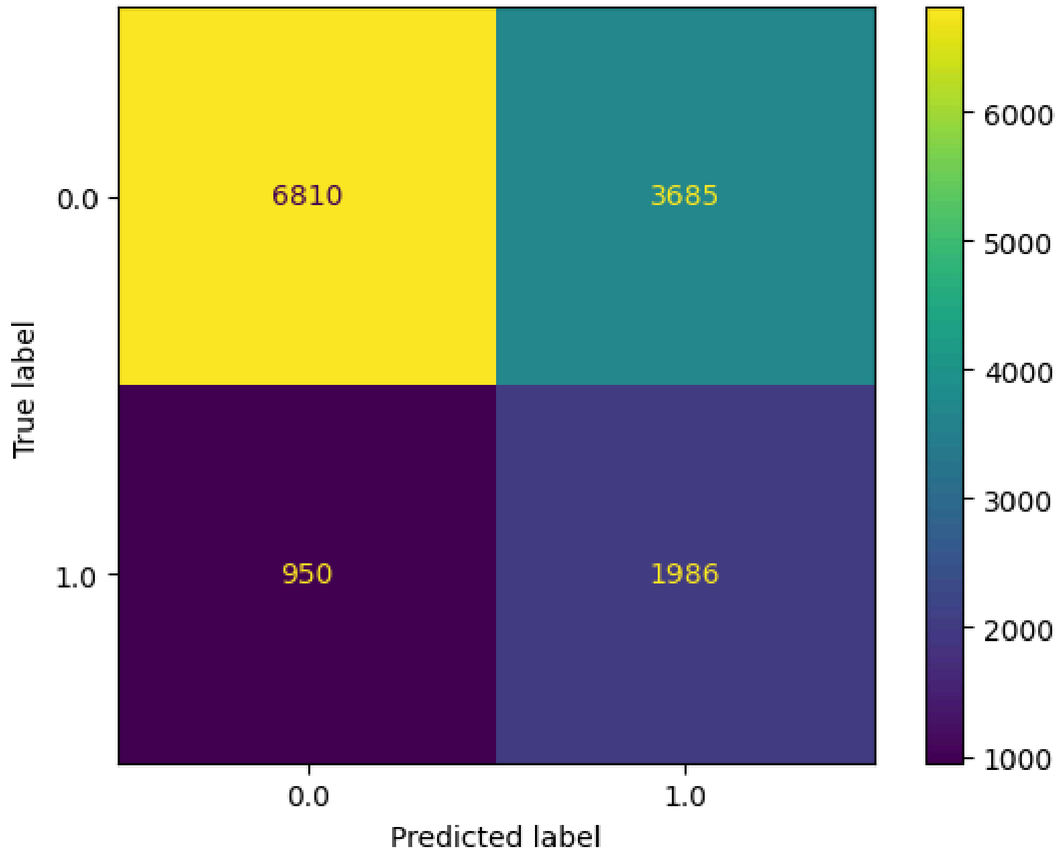
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.93 | 0.94 | 0.93 | 10495 |
| 1.0 | 0.77 | 0.76 | 0.76 | 2936 |
| accuracy | | | 0.90 | 13431 |
| macro avg | 0.85 | 0.85 | 0.85 | 13431 |
| weighted avg | 0.90 | 0.90 | 0.90 | 13431 |

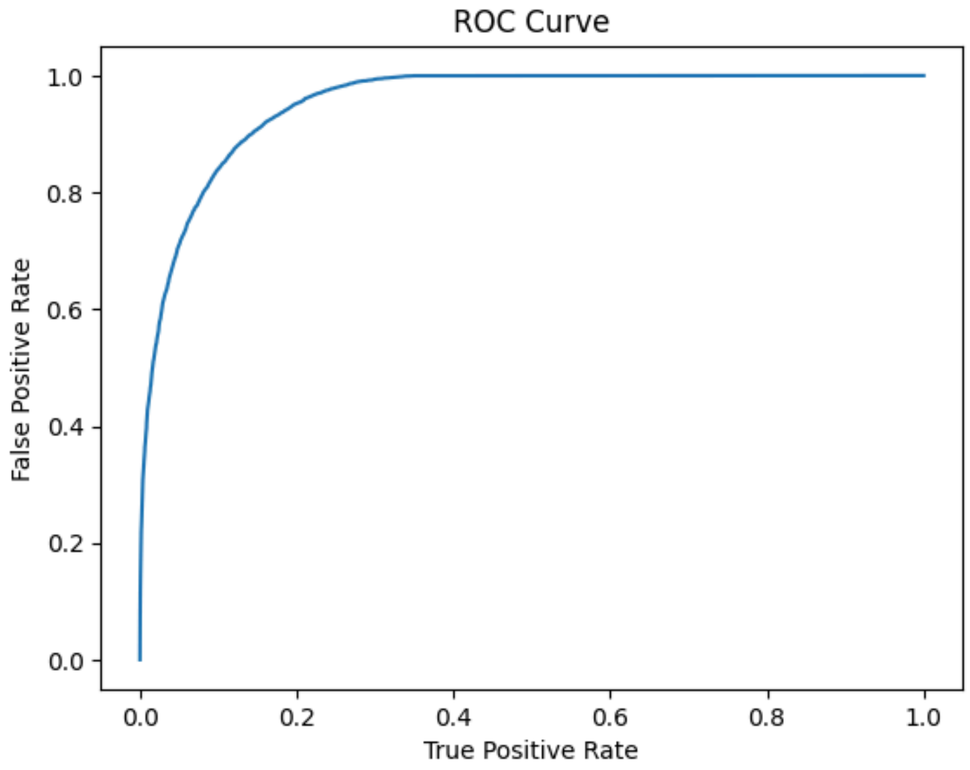


# PERFORMANCE: NAIVE BAYES

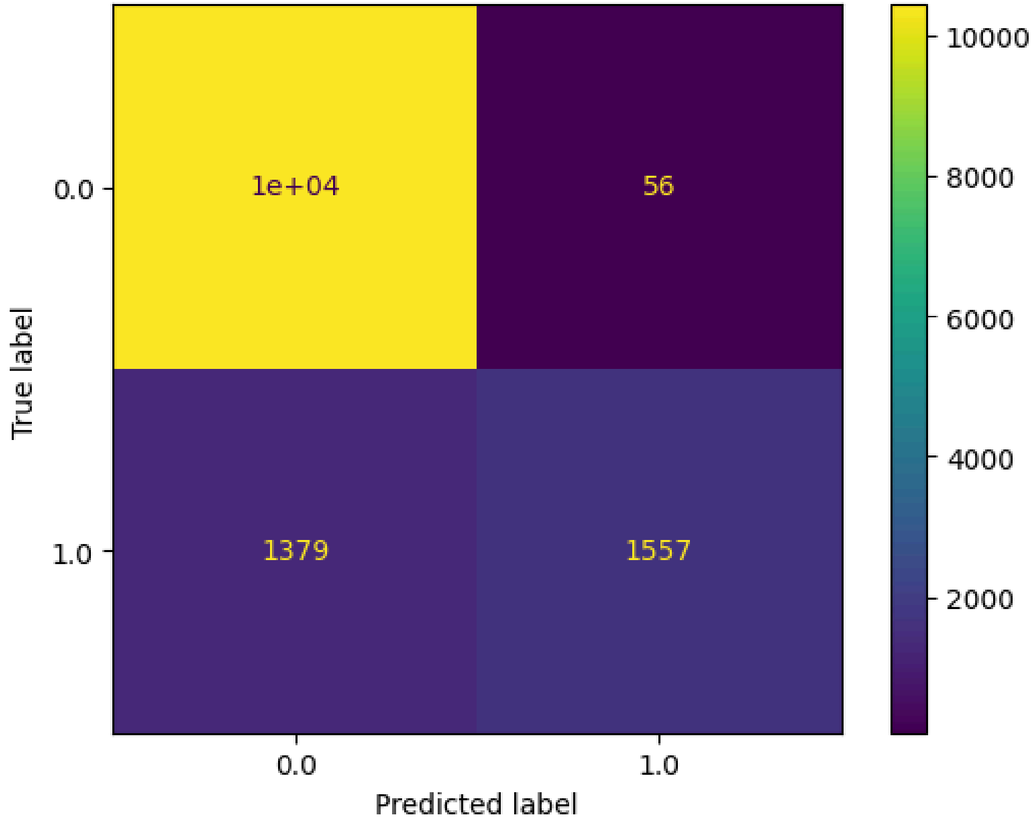
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.88 | 0.65 | 0.75 | 10495 |
| 1.0 | 0.35 | 0.68 | 0.46 | 2936 |
| accuracy | | | 0.65 | 13431 |
| macro avg | 0.61 | 0.66 | 0.60 | 13431 |
| weighted avg | 0.76 | 0.65 | 0.68 | 13431 |

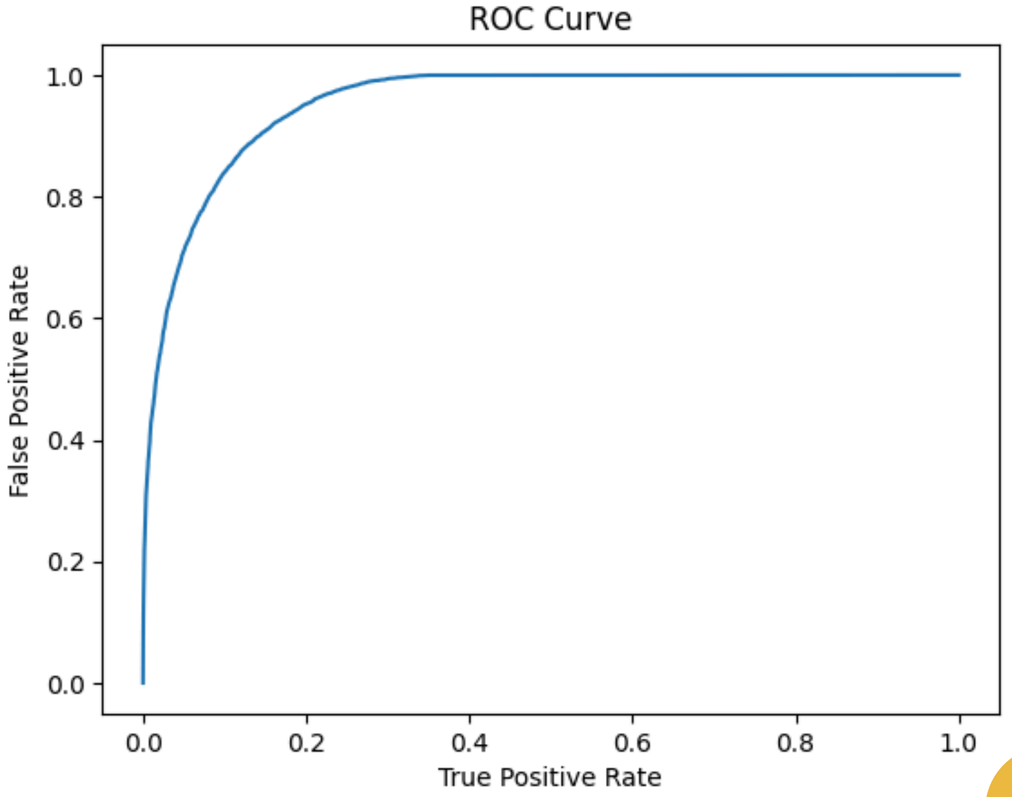


# PERFORMANCE: RANDOM FOREST

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0.0 | 0.88 | 0.99 | 0.94 | 10495 |
| 1.0 | 0.97 | 0.53 | 0.68 | 2936 |
| accuracy | | | 0.89 | 13431 |
| macro avg | 0.92 | 0.76 | 0.81 | 13431 |
| weighted avg | 0.90 | 0.89 | 0.88 | 13431 |



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# THANK YOU

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